Question	Answer	Marks
7(a)	Photoelectric Effect	**A1**
7(b)	$\frac{hc}{\lambda} = \phi + K$ $\frac{(6.63 \times 10^{-34})(3 \times 10^{8})}{(500 \times 10^{-9})} = (1.6 \times 10^{-19})(1.0) + K$ $K = 2.38 \times 10^{-19} J = 1.49\, eV$	**M1** **A1**
7(c)	$p = \sqrt{2mK}$ $p = \sqrt{2(9.11 \times 10^{-31})(2.38 \times 10^{-19})} = 6.58 \times 10^{-25}\, Ns$	**M1** **A1**
7(d)	$P = \frac{N\left(\frac{hc}{\lambda}\right)}{t}$ $25 \times 10^{-6} = \left(\frac{N}{t}\right)\frac{(6.63 \times 10^{-34})(3 \times 10^{8})}{(500 \times 10^{-9})}$ Number of incoming photons per unit time, $\frac{N}{t} = 6.285 \times 10^{13}$ Number of electron ejected per unit time $= (0.2)\left(\frac{N}{t}\right) = 1.257 \times 10^{13}$ Electron current, $I = (1.6 \times 10^{-19})(1.257 \times 10^{13}) = 2.01 \times 10^{-6}$ A	**M1** **M1** **A1**
7(e)	The metal must able to at least eject electrons from the least energetic photons. $\phi = \frac{hc}{\lambda}$ $\phi = \frac{(6.63 \times 10^{-34})(3 \times 10^{8})}{(700 \times 10^{-9})} = 2.84 \times 10^{-19} = 1.78\, eV$	**M1** **A1**
	Max Marks	**10**

Section B

Setter: Caleb Marker:		
Question	**Answer**	**Marks**
8(a)(i)	When two or more waves overlap (meet)	**B1**